Unit 2:- Ordinary Differentiation Equation:

Topics:

Linear and nonlinear differential equation, exact first order differential equations, Integrating factors.

Problems:

- Exercise 4.4 Q.1 to 45 (page 4.24, J& I)

Topics:

Solution of higher order homogeneous linear differential equations with constant coefficients.: Real and distinct roots, Real Multiple roots, simple complex roots, multiple complex roots.

Problems:

- Exercise 5.3, Q. 1 to 26. (page 5.25, J&I)
- Example 5.3, Q. 33 to 45 (page 5.25, J&I)

Topics:

Solution of non-homogeneous linear differential equations with constant coefficients.: Operator methods for finding particular integrals (page 5.44), solution of Euler Cauchy equation (page 5.37), Method of variation of parameters (5.27).

Problems:

- Exercise 5.7, Q. 1 to 50. (page 5.54, J&I)
- Exercise 5.6, Q. 1 to 45. (page 5.43, J&I)
- Exercise 5.4, Q. 1 to 12(page 5.32, J&I)

Topics:

Simultaneous linear differential equation: (page 381, CP)

Problems:

- Exercise 75 (page 381, CP)

Topics:

Applications to physical problems: Growth, decay and mixing problems (page 387, CP), Motion of particle (page 390, CP), Electrical circuits (page 392, CP).

Problems:

(Dr Sabiha Tabassum)
Course Incharge